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12 February 1976
SUPERSEDING
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19 August 1970

MILITARY SPECIFICATION

# REPAIR AND REHABILITATION OF RADAR SETS: GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers requirements for repair and rehabilitation of radar sets on site or in depot storage, ground mobile or transportable and land based search or control systems.

## 2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

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| MIL-E-1 | - Electron Tube, General Specification For. |
| MIL-T-152 | - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-E-4158 | - Electronic Equipment, Ground; General Requirements for. |
| MIL-S-6872 | - Soldering Process, General Specification for. |
| MIL-P-15024 | - Plate, Tags and Bands for Identification of Equipment. |
| MIL-E-16400 | - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification For. |
| MIL-E-17555 | - Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of. |
| MIL-I-45208 | - Inspection System Requirements. |



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STANDARDS

MILITARY

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| MIL-STD-129 | - Marking for Shipment And Storage. |
| MIL-STD-202 | - Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment. |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.1 General. The contractor shall perform all repair, calibration, performance testing, preparation for storage and shipment of equipment to the destinations specified in the contract or purchase order. Upon receipt of any complete system, assembly, subassembly and components thereof, the contractor shall conduct an inventory to determine completeness of the particular radar system. A complete listing shall be made of all items to be replaced or repaired for each major unit and each of its subassemblies. Upon completion of the listing, disassemble all major assemblies to the lowest possible allowable subassembly level. Repair and rehabilitation shall then be performed on all items (components and subassemblies as well as the major assemblies) which comprise the complete radar system. Any component of the particular radar system failing to meet the requirements specified herein shall be rejected and the contractor shall take such action as necessary to correct the deficiency. Disassembly as necessary will be made to insure compliance with this specification. Maintenance tools, specialized items, and test equipment required in the performance of the repair and rehabilitation shall be provided by the contractor.

3.1.1 Inventory. All items determined to be missing as a result of inventory shall be listed and submitted to the contracting officer. The Government shall determine if the missing items are to be contractor furnished or are to be furnished by the Government as Government-furnished material (GFM). As items are furnished as GFM or are manufactured/ purchased by the contractor, they shall be added to the inventory list required in 3.1.

3.1.2 Publications. All peculiar publications identified and associated with the particular radar system shall be provided as GFM.

3.1.3 Modification. A notation shall be made of all authorized modifications incorporated in the system. Authorized modifications required, but noted as not incorporated upon receipt of radar system, shall be provided to the contractor as GFM, for installation as a



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separate requirement, if available. If not available as GFM, the Government may elect to have the contractor furnish the required modifications under the line item in the contract identified as repair parts and materials.

3.1.4 Damaged GFM. If the GFM is received in a damaged condition, a DD Form 6, Package and Handling Deficiencies Report, must be submitted immediately. Any GFM damaged by the contractor through negligence in performance of repair and rehabilitation shall be replaced or repaired by the contractor at no additional expense to the Government.

3.3 Material. All parts or material used for replacement shall meet or exceed the performance specification of the original parts, shall be directly interchangeable with the original part, and shall comply with MIL-E-4158.

*3.2.1 Material and processes. Material and processes shall meet or exceed the requirements of MIL-I-45208 and MIL-STD-454; however, this specification shall govern in all respects where conflicts occur.

3.3 Repair. All defects shall be corrected. Repair, adjustment, alignment and calibration shall be conducted in accordance with the requirements of this specification. All test equipment and standards used during repair, rehabilitation and testing shall show a calibration date not to exceed a period of 6 months and have a traceability to the National Bureau of Standards.

3.3.1 Parts replacement. Worn or damaged parts shall be repaired whenever practicable. Parts shall be replaced only when repair is considered uneconomical. Economical repair is considered to be that repair which can be accomplished within 65 percent of the current unit acquisition price. Determination of economical repair versus replacement will be made by the contracting officer or by his duly authorized representative.

*3.3.1.1 Parts replacement, Government Furnished Property. To preclude delays in performance which could occur due to acts of God, defaults of subcontractors or causes beyond the contract and without the fault or negligence of the contractor, the Government may furnish the contractor Government Furnished Property (i.e., parts, components, radards, etc.). Radars furnished may be cannibalized to the extent authorized by the contracting officer. Should cannibalization be authorized, the contractor shall replace the parts or components used and shall place the radar in the condition as originally furnished or as otherwise directed by the contracting officer.

3.3.2 Bearings. All bearings shall be roll checked. Those determined defective shall be replaced.



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3.3.3 Gears. All gears, and gear combinations shall be checked for level wear, deterioration, slack and back-lash excessiveness. Engineering evaluation shall determine if any corrective action is required.

3.3.4 Structural components. All structural damage, either hardware, supporting, or assembly shall be repaired as required. This includes filling of dents and holes, with reshaping to appropriate contour.

3.3.5 Cables. All cables, coaxial or multiconductor, shall be repaired as required. All cables shall be cleaned to the maximum degree of practicability. All cables shall be tested by continuity and resistance checks.

3.3.5.1 Connectors. All irreparable cable connectors shall be replaced. All other connectors shall be repaired as required.

3.3.5.2 Sheathing. Damaged sheathing may be repaired by use of shrink tubing.

3.3.5.3 Ground cable. Ground cables may be spliced for repair if needed.

3.3.5.4 Lengths. All cables may be shortened by 10 percent to effect repairs, except in instances where minimum cable lengths are mandatory and might be affected.

3.3.5.5 Markers. Cable sleeve markers shall be made legible by remarkings, or replacing markers without removing connectors.

3.3.6 Waveguide. All waveguides and waveguide runs shall be disassembled for detail inspection. All waveguides shall be cleaned, and all waveguide gaskets shall be replaced. All nicks on the inside diameter of the waveguide flanges shall be removed by hand, using grade F, No. 40, abrasive.

3.3.7 Radio frequency (rf) switches. All waveguide rf coaxial switches shall be inspected, cleaned and tested electrically and mechanically as a separate item.

*3.3.8 Electron tubes. All electron tubes shall be tested for proper performance. Any tube which cannot meet 80 percent of its transconductance/emission rating, or has an internal element defect shall be replaced. All defective tubes shall be replaced with tubes of the same type as the original and shall comply with MIL-E-1.

3.3.9 Fabric bags. All tool bags, cable bags, transit bags, etc., which are made of fabric shall be patched or repaired as required to make them serviceable.



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3.3.10 Overall cleaning. The interior and exterior of all cases, cabinets and structures, and all assemblies, shall be cleaned to the maximum degree of practicability. Components exhibiting signs of corrosion or discoloration which do not affect performance shall be treated to retard further effect, but shall not require replacement.

3.4 Painting. The exterior of all external equipment susceptible to weather environment shall be repainted in accordance with type A of MIL-T-704. The color shall match Marine Corps green, color No. 23 (see 6.3). The interiors of shelters or interior of equipment cabinets shall be repainted or touched up with paint matching the original color.

3.4.1 Stenciling. All stenciling shall be masked where repainting is required. Any stenciling required may be made by hand, except for those describing modifications.

3.5 Identification plates. All identification plates shall have correct and complete data. Any defaced or unreadable plates shall be replaced in accordance with MIL-P-15024.

3.6 Workmanship. The equipment shall be repaired in a thoroughly professional manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, working of parts and assemblies, cleaning and painting of equipment, and alignment procedures conforming to MIL-T-152, MIL-S-6872 and MIL-E-16400 as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection, test, and acceptance. Upon completion of the repair and rehabilitation, all systems shall be operationally and electrically inspected and tested at the acceptance site in accordance with the inspection and test procedures and criteria as set forth in this specification including GFM publications, manuals, etc.

4.2.1 Acceptance. Acceptance shall be by the Government inspector or an agent of the procuring activity. Failure of any system or subsystem to meet the minimum performance characteristics specified herein shall be cause for rejection of the end item. End item shall be reworked to the extent necessary to correct the deficiency.



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*4.2.1.1 AN/TPQ-10/10A acceptance testing. For repair and rehabilitation of the Radar Course Directing Central, AN/TPQ-10/10A, only, acceptance shall include a demonstration of bombing accuracy. Bombing accuracy shall be (1) the ability in an automatic mode (ground control bombing system) to deliver 50 percent of the bombs within a 50-meter circular area of a prescribed target (a minimum of 30 bombs shall be dropped involving multiple runs); and (2) the ability in the manual mode, to deliver 50 percent of the bombs within a 100-meter circular area of a prescribed target (a minimum of 30 bombs shall be dropped involving multiple runs); or (3) as determined by the Government inspector, or an agent of the procuring activity, any combination of 30 bombs meeting the above two modes shall be acceptable. Failure to meet this acceptance criteria shall be subject to approval in writing of a deviation, on-site, by the project officer or his duly authorized representative.

4.2.2 Notification. The final inspection and testing will be witnessed by the Government inspector or an agent of the procuring activity. The contractor shall notify the contracting officer not later than 10 days prior to the scheduled testing.

4.2.3 Inspection check list. A proposed inspection and test check list including test procedures shall be submitted to the contracting officer for review and approval not later than 30 days prior to the scheduled testing.

4.3 Inspection. The list required in 3.1 shall be utilized as the basis for inspection to determine completeness of the radar system.

4.4 Test methods and procedures. Requirements set forth in Section 3 of this specification shall be met utilizing procedures referenced in the following paragraphs.

4.4.1 Mechanical inspection tests. The radar set shall be given a mechanical and visual inspection and test to determine that the quality of all material workmanship is in compliance with the requirements of this specification. Particular attention shall be given the following:

(a) Nameplates, identification markings and labels.

(b) Ease of operation of gears, adjustable and sliding parts, thumb screws, controls and switches.

(c) Finishes.

(d) Welded joints and structural members.



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(e) Solder joints.

(f) Fit of components in their respective positions, including replacement parts.

(g) Check of mounting means.

(h) Check of lubrication and rust prevention.

(i) Check of safety features and interlocks.

(j) Loose fastenings and securing devices and parts.

(k) Proper replacement parts and material.

(l) Cables, cable runs including plug, connectors and receptacles.

(m) Grounding connections.

(n) Any and all other visual defects.

4.4.2 Electrical system test. Each radar set shall be given a complete electrical system test to ensure that the overall performance and operation conform to the requirements of this specification and MIL-STD-202. These tests shall include those system tests, as contained in the particular technical publication, which pertain to the particular radar.

4.4.3 Unit tests. Upon completion of repair and rehabilitation, the major assemblies shall be reassembled, and subjected to factory or operational unit test. This test shall:

(a) Incorporate, where applicable, alignments and checkout of wired cards, switches, relays, etc.

(b) Incorporate, where applicable, checkout of motors, pumps, hydraulic systems, drives, etc.

(c) Be instituted to assure proper operation and performance of the refurbished radar system, and shall be conducted with sufficient detail to insure the detection of all malfunctions and misalignments. Repair and readjustment shall be instituted when necessary.

4.4.4 System checkout and acceptance. Upon completion of unit tests, the radar system shall be erected in its characteristic configuration for full system checkout and acceptance, to include, but not to be limited to, the following: radomes, pulse tanks, rotary joints, coolant assemblies, computers, data boxes, antenna assemblies, high voltage slip rings, etc.



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4.5 Final tests. Final tests shall include the following:

(a) A complete system alignment and test shall be conducted to insure proper operation and performance of the radar system prior to final acceptance.

(b) System tests shall be conducted with the equipment completely assembled in its operational configuration.

(c) All operational characteristics shall be equal to or better than those specified for acceptance when the equipment was initially procured, with the exception that where authorized modifications have changed the original specifications, the modified specifications shall apply.

(d) Peculiar test requirements will apply, as a separate contract requirement in addition to the accomplishment of the repair and rehabilitation procedure, for the specific radar system to be refurbished.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. This specification is intended to provide requirements for the repair and rehabilitation of radar sets used for the detection or control of aircraft at ranges up to 300 nautical miles at altitudes up to 100,000 feet or for repair and rehabilitation of the Radar Course Directing Central, AN/TPQ-10/10A, used for acquisition and direction of aircraft on a preplanned course for bombing purposes.

*6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

*6.2.1 Procurement requirements.

(a) Title, number and date of this specification.

(b) Levels of preservation, packaging and packing (see 5.1).

*6.2.2 Contract data requirements. Data conforming to Data Item



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Descriptions DI-R-4802 and DI-T-4909 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

*6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps, Headquarters U. S. Marine Corps (Code LMO-1), Washington, D. C. 20380.

6.4 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship of the last previous issue.

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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